

CareerMatch AI – Data Analyst Role Guide

Synthetic/Generalized Career Knowledge – Hackathon Prototype

1. Role Overview

A Data Analyst turns raw business data into clear insights that support decision-making. The role emphasizes querying, cleaning, and visualizing data more than building predictive models. This is generalized educational content for a hackathon prototype.

2. Responsibilities

- Extract data from databases using SQL
- Clean and organize data for analysis
- Build reports and dashboards for stakeholders
- Identify trends, patterns, and anomalies in business data
- Present findings in a clear, actionable way

3. Required Skills

SQL

What it is: A query language used to read, filter, and aggregate data stored in relational databases.

Why it matters: Most real-world data lives in databases, so extracting and shaping it correctly is a foundational skill for any data-related role.

How it's used in this role: Used to pull structured datasets from a company database before analysis or model training.

Beginner expectation: Can write SELECT, WHERE, GROUP BY, and simple JOIN queries.

Excel

What it is: A spreadsheet application widely used for data entry, calculation, and basic analysis.

Why it matters: It remains a common first tool for business data analysis and reporting.

How it's used in this role: Used for quick analysis and reporting before or alongside programmatic tools.

Beginner expectation: Can use formulas, sorting, and filtering on a dataset.

Python

What it is: A general-purpose, beginner-friendly programming language widely used for data work, automation, and AI development.

Why it matters: It is the primary language for machine learning, data analysis, and AI tooling because of its simple syntax and large ecosystem of libraries.

How it's used in this role: Used to load data, build models, write automation scripts, and connect components of an AI pipeline.

Beginner expectation: Can write scripts using variables, loops, conditionals, functions, and lists/dictionaries.

Pandas

What it is: A Python library for loading, cleaning, and manipulating tabular data.

Why it matters: It is the standard tool for everyday data wrangling tasks before analysis or modeling.

How it's used in this role: Used to clean and reshape a dataset before exploratory analysis or model training.

Beginner expectation: Can load a CSV, filter rows, and compute simple summary statistics.

Data Cleaning

What it is: The process of fixing or removing incorrect, incomplete, or inconsistent data.

Why it matters: Poor quality data leads to unreliable models, regardless of how advanced the algorithm is.

How it's used in this role: Used as the first practical step in nearly every data project.

Beginner expectation: Can handle missing values, duplicates, and obvious formatting errors.

Statistics

What it is: The study of collecting, analyzing, and interpreting numerical data.

Why it matters: Machine learning is built on statistical reasoning, from describing data to validating model results.

How it's used in this role: Used to summarize data and validate whether observed patterns are meaningful.

Beginner expectation: Understands mean, median, mode, standard deviation, and basic distributions.

Data Visualization

What it is: The practice of representing data visually through charts and graphs.

Why it matters: It communicates findings clearly to both technical and non-technical audiences.

How it's used in this role: Used to present analysis results and support data-driven decisions.

Beginner expectation: Can create basic charts to summarize a dataset.

4. BI Tools

Power BI

What it is: A business intelligence tool used to build interactive dashboards and reports.

Why it matters: Organizations rely on BI dashboards to monitor metrics without needing to write code.

How it's used in this role: Used to present business metrics to stakeholders in a visual, interactive format.

Beginner expectation: Can build a simple dashboard from a dataset.

Tableau

What it is: A business intelligence tool for creating interactive data visualizations and dashboards.

Why it matters: It is widely used in industry for turning data into visual, shareable insights.

How it's used in this role: Used to build dashboards that communicate analysis results to a broader audience.

Beginner expectation: Can build simple charts and dashboards.

5. Important SQL Topics

- **SELECT:** Retrieves specific columns of data from one or more tables.
- **WHERE:** Filters rows based on a condition before they are returned.
- **GROUP BY:** Groups rows that share a value so aggregate calculations can be applied to each group.
- **HAVING:** Filters groups after aggregation, similar to WHERE but for grouped data.
- **JOIN:** Combines rows from two or more tables based on a related column.
- **Subqueries:** A query nested inside another query, used to break complex logic into steps.
- **Aggregate Functions:** Functions like SUM, AVG, COUNT, MIN, and MAX that summarize a group of rows.

6. Education

Common backgrounds include Business Administration, Statistics, Computer Science, Economics, or Commerce, along with self-taught SQL, Excel, and visualization skills.

7. Fresher Expectations

- Can write SQL queries involving filtering, grouping, and joins
- Comfortable building simple dashboards or reports
- Can clean a messy spreadsheet and identify data quality issues
- Can explain a trend or finding in plain business language
- These expectations describe a general, entry-level readiness bar used only for hackathon prototyping. They are illustrative and not tied to any real company's actual hiring bar, vacancy, or deadline.

8. High/Medium/Low Priority Skills

High Priority	Medium Priority	Low Priority
• SQL • Excel • Data Cleaning	• Python • Pandas • Data Visualization	• Power BI • Tableau • Statistics

9. Common Skill Gaps

- Comfortable with Excel but limited SQL query experience
- Difficulty joining multiple tables correctly
- Dashboards that show data without a clear takeaway
- Limited practice cleaning genuinely messy, real-world data

10. Portfolio Projects

1. Local Transport Delay Analysis

Problem: Analyze public transport data to identify common causes of delay.

Skills used: SQL, Data Cleaning, Data Visualization

Difficulty: Beginner

2. Small Business Sales Dashboard

Problem: Build an interactive dashboard summarizing sales performance over time.

Skills used: SQL, Power BI or Tableau

Difficulty: Beginner to Intermediate

11. Interview Topics

- SQL query writing and optimization
- Data cleaning techniques
- Dashboard and report design principles
- Basic statistics for interpreting business metrics

12. Interview Questions

- 1 Write a query to find the second-highest value in a column.
- 2 How would you handle duplicate records in a dataset?
- 3 What is the difference between WHERE and HAVING?
- 4 How would you design a dashboard for a sales team?
- 5 Explain a time you found an unexpected insight in data.

13. Learning Roadmap

- 1 Excel fundamentals
- 2 SQL fundamentals
- 3 Python and Pandas basics
- 4 Data cleaning practice
- 5 Data visualization
- 6 BI tool basics (Power BI or Tableau)

14. RAG Retrieval Keywords

data analyst, SQL for data analysis, Power BI, Tableau, dashboard design, entry-level data analyst

Document Metadata

Job ID	DA001
Role	Data Analyst
Category	Data Analytics
Experience Level	Entry Level
Data Status	Synthetic/Generalized